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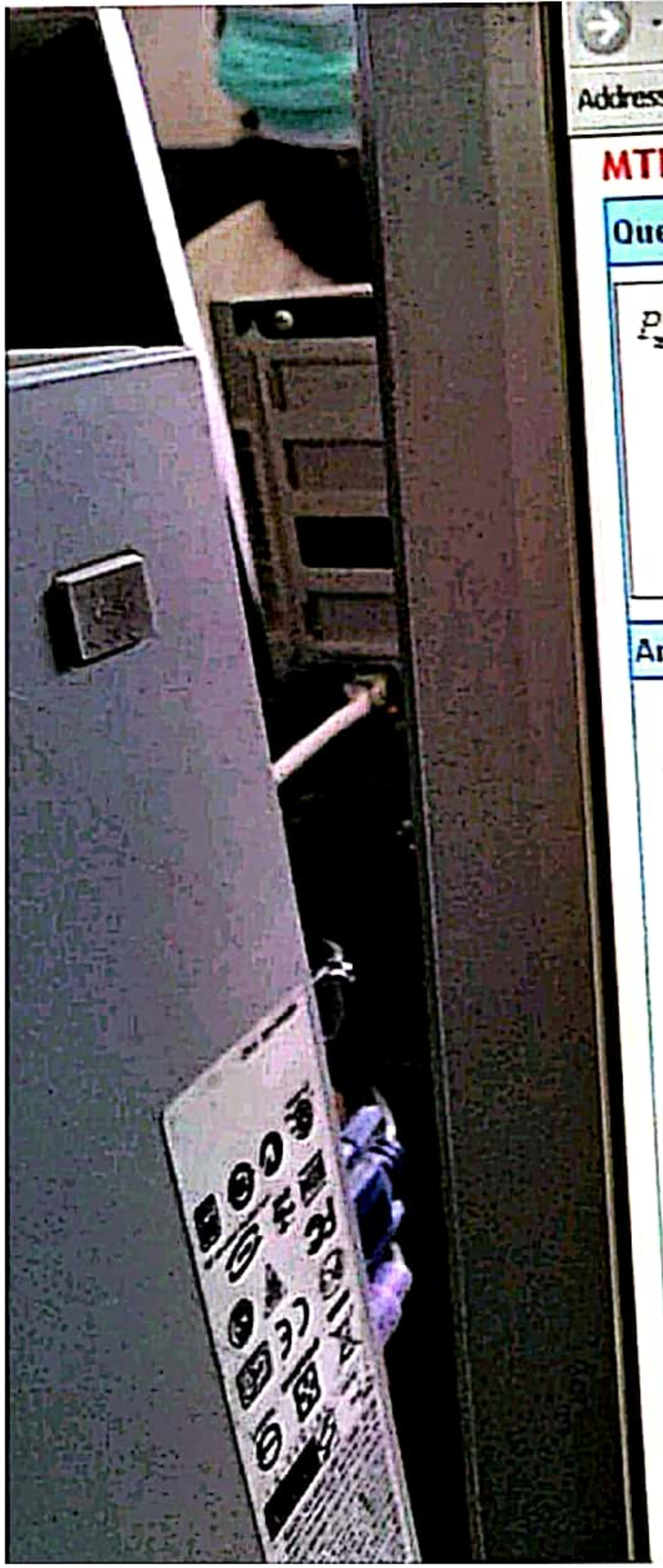
MTH641 Functional Analysis

Question No : 35 of 41

Prove that if a set M is closed in X then it forms a complete subspace of X

Answer (Please click here to Add Answer)

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MTH641 Functional Analysis

Question No : 28 of 41

P_n is the collection of all.....

Answer (Please select your correct option)

- Polynomials of degree greater then equal to 'n'
- Polynomials of degree less then equal to 'n'
- Polynomials of degree equal to 'n'
- Polynomials of degree less then 'n'

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Question Summary : (Attempted

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MTH641 Functional Analysis

Question No : 29 of 41

The pair $(V, \langle .,. \rangle)$ is called .....space

Answer (Please select your correct option)

- Norm
- Metric
- Inner product
- Vector

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MTH641 Functional Analysis

Question No : 16 of 41

Which of the following is a condition of an inner product space?

Answer (Please select your correct option)

- $\langle \alpha x, y \rangle = \langle x, \alpha y \rangle$
- $\langle \alpha x + y, z \rangle = \langle x, z \rangle + \alpha \langle y, z \rangle$
- $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$ ✓
- $\langle x, y \rangle = \overline{\langle x, y \rangle}$

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 23 of 41

Which of the following is not true for the space $C[a,b]$ with norm defined by $\|x\| = \max_{t\in[a,b]} |x(t)|$?

Answer (Please select your correct option)

- It is a complete metric space
- It is a complete norm space
- It is a complete inner product space ✓
- It is a vector space

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 5 of 41

A _____ Y of a Banach space X is _____ if and only if the set Y is closed in X

Answer (Please select your correct option)

- subspace , complete
- space, complete
- subspace, open
- space, open

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Question Summary : (Attempted

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Question No : 20 of 41

Every metric space is a not _____ space.

Answer (Please select your correct option)

- a normed ✓
- a topological
- both of these
- None of these

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 39 of 41

Show that canonical mapping is linear.

Answer (Please click here to Add Answer)

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MTH641 Functional Analysis

Question No : 17 of 41

Which of the following is a condition of an inner product space?

Answer (Please select your correct option)

- $\langle x,x\rangle \geq 0$ ✓
- $\langle \alpha x+y,z\rangle = \langle x,z\rangle + \alpha\langle y,z\rangle$
- $\langle \alpha x,y\rangle = \langle x,\alpha y\rangle$
- $\langle x,y\rangle = \overline{\langle x,y\rangle}$

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Question Summary : (Attempted Question)

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Question No : 37 of 41

Define norm and inner product for which the space R^n is an inner product space.

Answer (Please click here to Add Answer)

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MTH641 Functional Analysis

Question No : 24 of 41

The space .....is not a Hilbert space

Answer (Please select your correct option)

- l^p with $p \neq 2$ ✓
- Euclidean space R^n
- Unitary Space C^n
- l^p with $p = 2$

l^p with $p \neq 2$

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MTH641 Functional Analysis

Question No : 30 of 41

Inner product is the generalization of .....

Answer (Please select your correct option)

- Cross product
- Dot product ✓ dot product
- Multiplication
- Distance function

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 32 of 41

Define linear functional

Answer (Please click here to Add Answer)

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MTH641 Functional Analysis

Question No : 18 of 41

Which of the following is true about the inner product space?

Answer (Please select your correct option)

- Every normed space is an inner product space.
- Every inner product space is a normed space. ✓
- Every inner product space is Hilbert space.
- None of these is true.

Every inner product space is a normed space

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 10 of 41

A linear functional f with domain $D(f)$ in a normed space is continuous if and only if f is _____.

Answer (Please select your correct option)

- continuous
- bounded
- linear
- non linear

bounded ✓

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 3 of 41

Given a point $x_0 \in X$ and a real number $r > 0$, which of the following defines a closed ball ?

Answer (Please select your correct option)

- $\bar{B}(x_0;r)=\{x\in X \mid d(x,x_0) < r\}$
- $\bar{B}(x_0;r)=\{x\in X \mid d(x,x_0) > r\}$
- $\bar{B}(x_0;r)=\{x\in X \mid d(x,x_0) \leq r\}$ ✓
- $\bar{B}(x_0;r)=\{x\in X \mid d(x,x_0) = r\}$

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 11 of 41

The space $C[a,b]$ is _____.

Answer (Please select your correct option)

- collection of all functions defined on $[a,b]$.
- collection of all continuous functions defined on (a,b)
- collection of all continuous functions defined on $[a,b]$ ✓
- collection of all functions defined on (a,b)

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

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Question No : 41 of 41

Marks: 5 (Budgeted Time Min)

Let H_1 and H_2 be Hilbert spaces, $S: H_1 \to H_2$ and $T: H_1 \to H_2$ be bounded linear operator and α any scalar. Then show that $(ST)^* = T^* S^*$ (assuming $H_2 = H_1$).

Answer (Please click here to Add Answer)

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 34 of 41

Define unitary operator.

Answer (Please click here to Add Answer)

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MTH641 Functional Analysis

Question No : 36 of 41 **Marks: 3 (Budgete**

Let $T : D(T) \to Y$ be a linear operator, where $D(T) \subseteq X$ and X, Y are normed spaces, then show that if T is continuous at a single point, it is continuous.

Answer (Please click here to Add Answer)

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MTH641 Functional Analysis

Question No : 12 of 41

Let X be an n-dimensional vector space and X^* be its dual space. Then _____.

Answer (Please select your correct option)

- $\dim X^* = \dim X = n$

  dim X^*= dim X=n ✓
- $\dim X = n$
- $\dim X^* = n$
- $\dim X^* = \dim X = m$

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 22 of 41

Which of the following is an example of an inner product space.

Answer (Please select your correct option)

- (a,b)
- $[a,b]$
- Space R^*

Space R* ✓

- None of these

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MTH641 Functional Analysis

Question No : 2 of 41

For any three points x,y and z in a metric space (X,d), the following property is called .....
$d(x,z) \leq d(x,y) + d(y,z)$

Answer (Please select your correct option)

- Trianguar inequality
- Triangular inequality ✓
- Rectangle inequality
- Holder's inequality
- Minkowski's inequality

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Question Summary : (Attempted: 2

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MTH641 Functional Analysis

Question No : 7 of 41

A metric space X is said to be compact if every sequence in X has a _____ subsequence.

Answer (Please select your correct option)

- Non decreasing
- Divergent
- Non increasing
- Convergent

convergent ✓

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 9 of 41

Let T be a bounded linear operator. Then the _____ is closed.

Answer (Please select your correct option)

- Banach space
- null space $N(T)$
- Hilbert space
- None of these

Null space N(T) ✓

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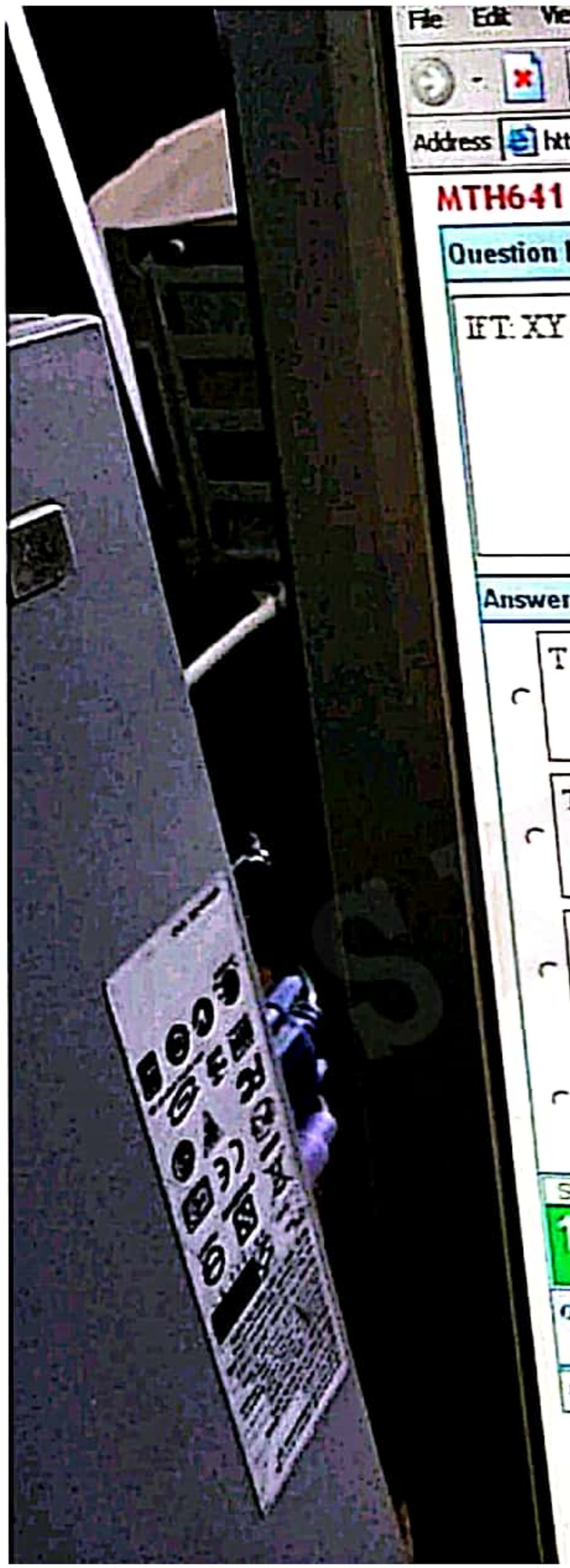
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MTH641 Functional Analysis

Question No : 14 of 41

IF T: XY is an Isometric Isomorphism between two norm spaces X and Y, then it implies that.....

Answer (Please select your correct option)

- T is bijective and norm preserving
- T is injective and norm preserving
- T is surjective and norm preserving
- T is norm preserving

T is bijective and norm preserving ✓

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 6 of 41 **Marks: 1 (Budgeted**

Let $\{x_1,...,x_2\}$ be .....set of vectors in a normed space X. Then there is a number $c > 0$ such that for every choice of scalars $\alpha_1,...,\alpha_2$, we have

$|\alpha_1 x_1 + ... + \alpha_2 x_2| \geq c(|\alpha_1| + ... + |\alpha_2|)$

Answer (Please select your correct option)

- linearly independent

Linearly independent ✓

- linearly dependent
- orthogonal
- orthonormal

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 21 of 41

Every complete normed space is _____ space.

Answer (Please select your correct option)

- Hilbert
- Banach ✓
- Unitary
- Euclidean

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MTH641 Functional Analysis

Question No : 13 of 41 **Marks:**

Let X be a finite dimensional vector space. If $x_0 \in X$ has the property that $f(x_0) = 0$ for all $f \in X^*$, then _____.

Answer (Please select your correct option)

- $x_0 = 0$ $x_o = 0$ ✓
- $x_0 = 0$
- $x_0 = 1$
- $x_0 = 1$

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 15 of 41

The dual space of a normed space X contains all ..... on X

Answer (Please select your correct option)

- Bounded linear functional

Bounded linear functional ✓

- Unbounded linear functional
- Bounded linear operator
- Unbounded linear operator

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 19 of 41

Every inner product space is a _____ space.

Answer (Please select your correct option)

- Metric
- Normed ✓
- Topological
- All of above

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MTH641 Functional Analysis

Question No : 33 of 41

Write generalized form of orthogonal system

Answer (Please click here to Add Answer)

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MTH641 Functional Analysis

Question No : 1 of 41

The property $d(x, y) \geq 0$ of a metric space is called.....

Answer (Please select your correct option)

- Non negativity ✓
- Reflexive property
- Symmetric property
- Triangle property

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Question Summary : (Attempted Question

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MTH641 Functional Analysis

Question No : 31 of 41

Show that the sequence $(x_n) = (1 + \frac{1}{n})$ is a Cauchy sequence with the usual metric defined on the set of real numbers.

Answer (Please click here to Add Answer)

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T and X, Y are normed spaces, then show that if T is continuous at a single point, t

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MTH641 Functional Analysis

Question No : 25 of 41 **Marks: 1 (Budge**

Isomorphism of an inner product space onto an inner product space over the same field is a .....linear operator that preserves inner product.

Answer (Please select your correct option)

- Bijective
- Injective
- Surjective
- None of these

Bijective ✓

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 4 of 41

Two metric spaces (X,d) and $(\bar{X},\bar{d})$ are said to be isometric if there exists ..... isometry of X onto $\bar{X}$

Answer (Please select your correct option)

- An injective
- A bijective
- A bijective ✓
- A surjective
- none of these

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 38 of 41

Let H_1 and H_2 be Hilbert spaces, $S: H_1 \to H_2$ and $T: H_1 \to H_2$ be bounded linear operator and α any scalar. Then show that

$\langle T^* y, x \rangle = \langle y, Tx \rangle \qquad (x \in H_1, y \in H_2)$.

Answer (Please click here to Add Answer)

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MTH641 Functional Analysis

Question No : 8 of 41 **Marks: 1 (Budgeted T**

Let $T : D(T) \to Y$ be a linear operator, where $D(T) \subseteq X$ and X, Y are normed spaces, then T is _____ if and only if T is bounded

Answer (Please select your correct option)

- continous
- linear
- bounded
- none of these

continous ✓

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

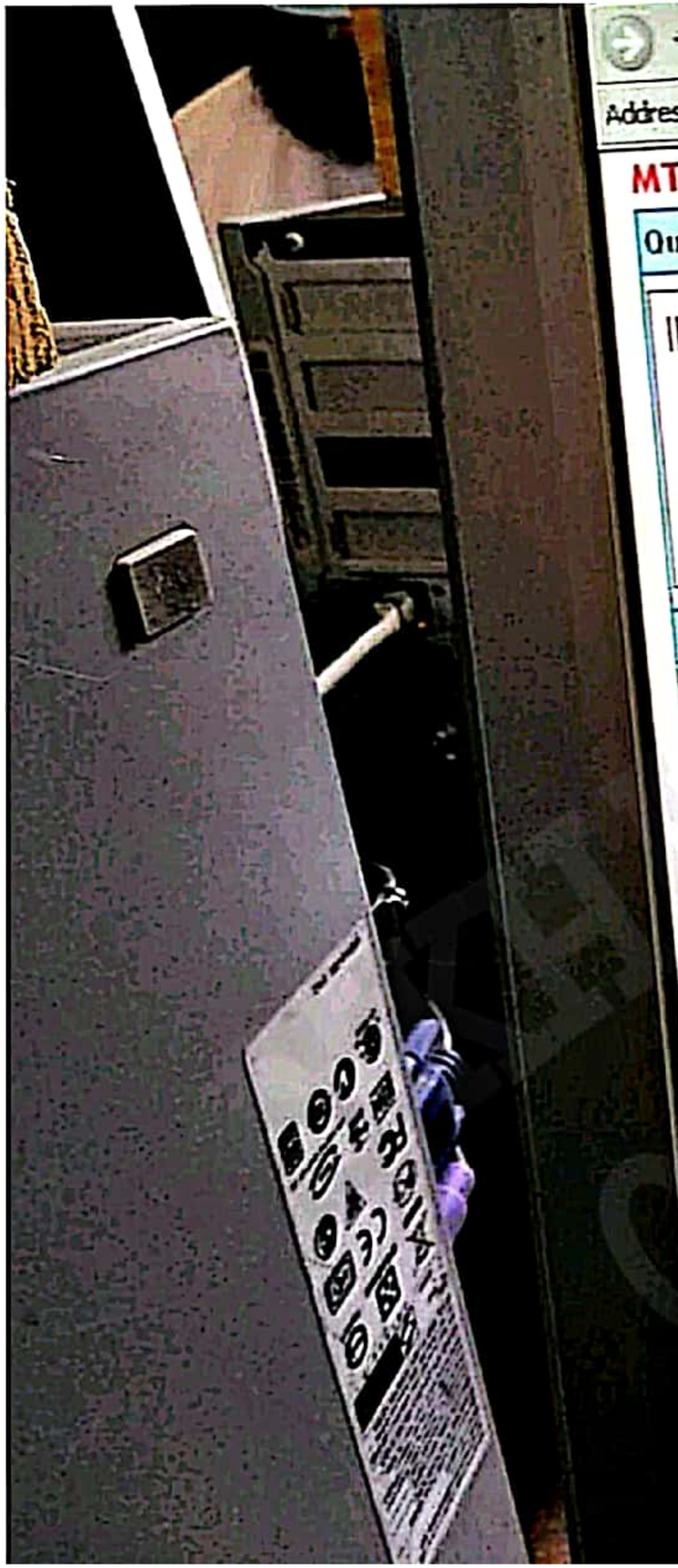
Question No : 40 of 41

Show that if $\{x_n\}, \{y_n\}$ are Cauchy sequences in V, then $\{x_n, y_n\}$ is a Cauchy sequence in F.

Answer (Please click here to Add Answer)

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MTH641 Functional Analysis

Question No : 26 of 41

$\| x+y \| \leq \| x \| + \| y \|$ is called.....

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Answer (Please select your correct option)

- Triangle inequality
- triangular inequality ✓
- Minkowski's inequality
- Holder's inequality
- Schwarz inequality

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Question Summary : (Attempted Question)

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MTH641 Functional Analysis

Question No : 27 of 41

For orthogonal elements x,y we have <x,y>=0 and the Pythagorean relation is .....

Answer (Please select your correct option)

- $\|x+y\|^2=2\|x\|^2+2\|y\|^2$
- $\|x+y\|^2=\|x\|^2-\|y\|^2$
- $\|x-y\|^2=\|x\|^2+\|y\|^2$
- $\|x+y\|^2=\|x\|^2+\|y\|^2$ ✓

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2$$

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Question Summary : (Attempted Question)

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